

Machine Learning

Lecture – 10

Unsupervised Learning

What is Cluster Analysis?

- Cluster: a collection of data objects
 - Similar to one another within the same cluster
 - Dissimilar to the objects in other clusters
- Cluster analysis
 - Grouping a set of data objects into clusters
- Clustering is unsupervised classification: no predefined classes
- Typical applications
 - As a stand-alone tool to get insight into data distribution
 - As a preprocessing step for other algorithms

Often used as a data analysis tool

Typical Applications of Clustering

- Clustering Documents:
 - Based on keyword occurrence frequency (e.g., Bag of Words features)
- Image Segmentation:
 - Compression, Object detection, etc.
- Marketing:
 - Help marketers discover distinct groups in their customer bases, and then use this knowledge to develop targeted marketing programs
- Insurance:
 - Identifying groups of motor insurance policy holders with a high average claim cost

What is a Good Clustering?

- A good clustering method will produce high quality clusters with
 - high within-class similarity
 - low between-class similarity
- Given the data, the quality of a clustering result depends on
 - similarity measure used
 - the method and its implementation.
- The quality of a clustering method is also measured by its ability to discover some or all of the hidden patterns.

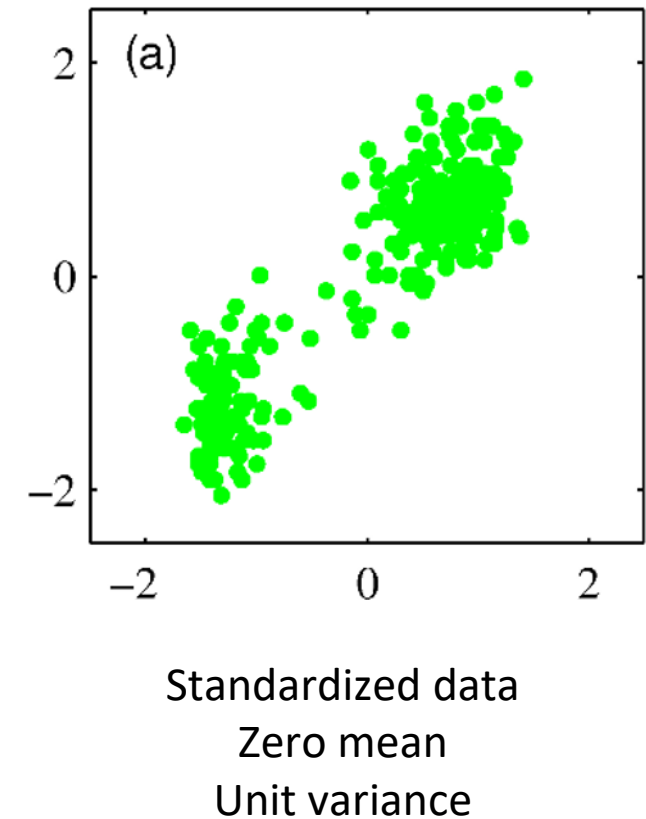
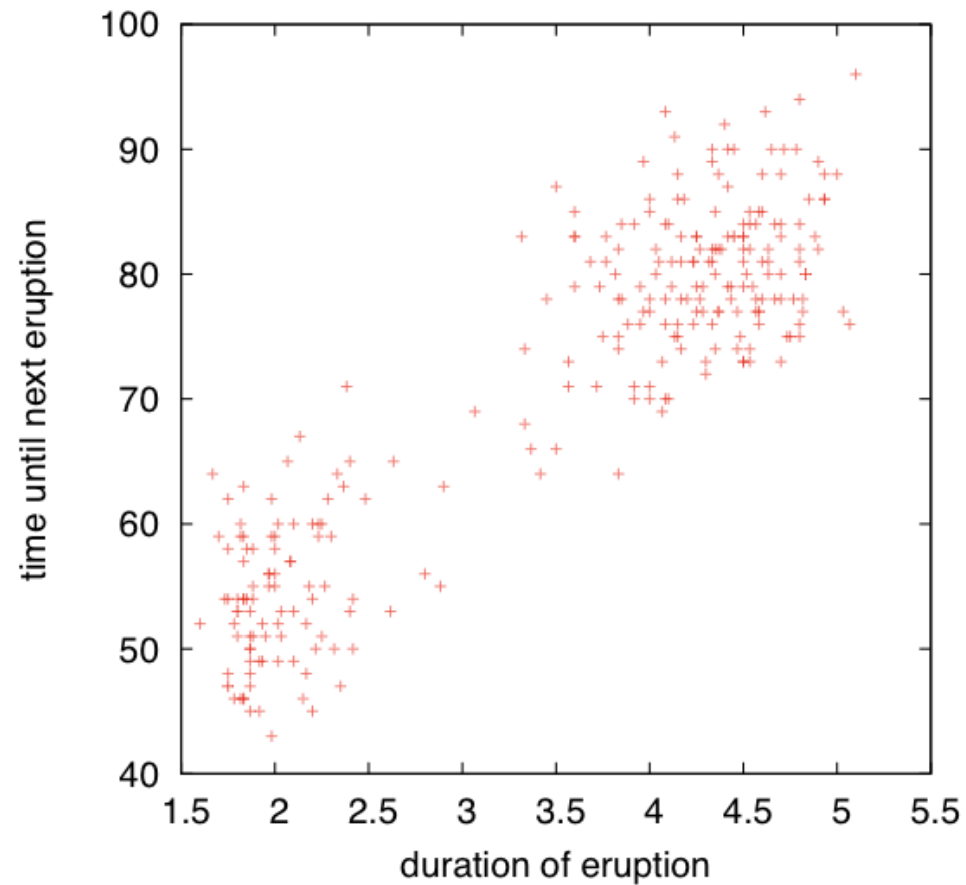
Different Clustering Paradigms

- Partitioning algorithms:
 - Construct various partitions and then evaluate them by some criterion, e.g., k-means and its variants
- Hierarchical algorithms:
 - Create a hierarchical decomposition of the set of data (or objects) using some criterion, e.g., single-linkage, complete-linkage, average-linkage
- Density-based:
 - based on connectivity and density functions, e.g., DBSCAN, OPTICS, Mean Shift Clustering
- Model-based:
 - A model is hypothesized for each of the clusters and the idea is to find the best fit of that model to each other, e.g., Mixture of Gaussians, Mixture of Binomials

Old Faithful data

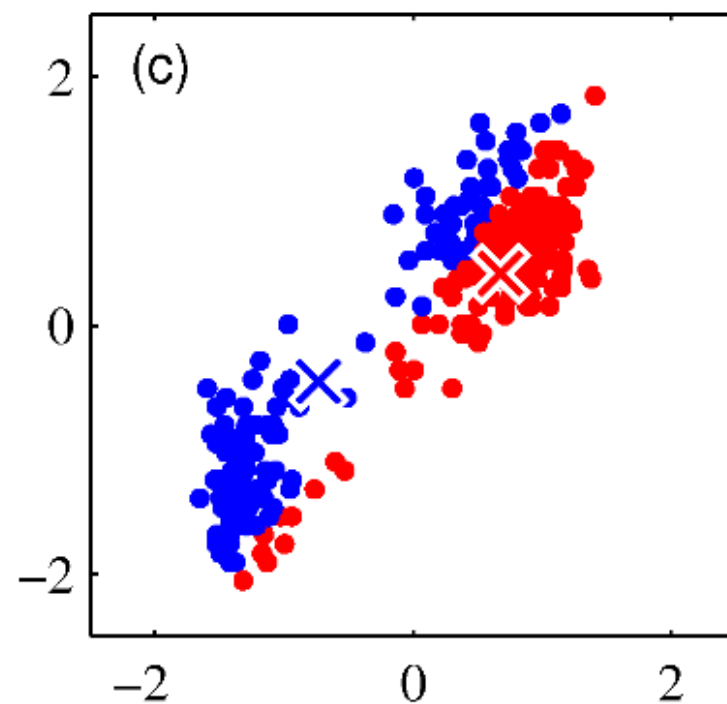
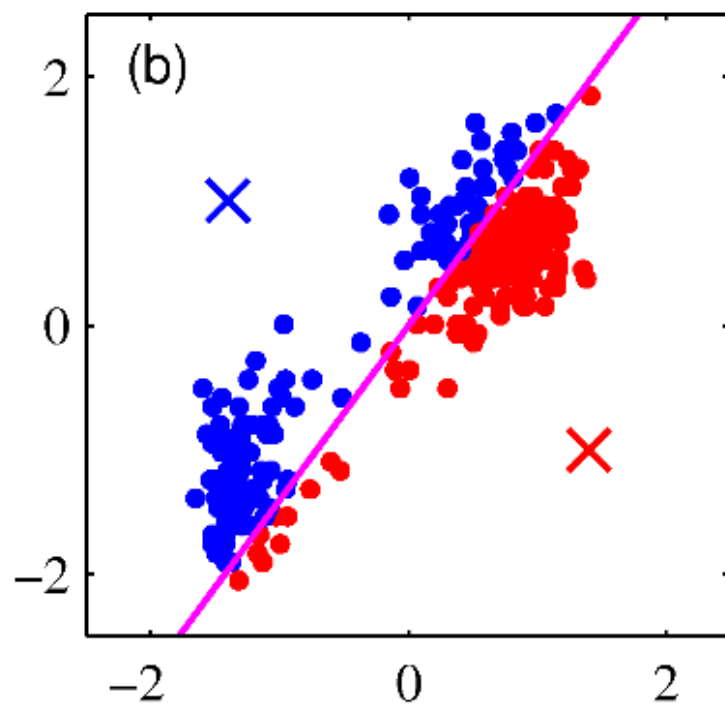
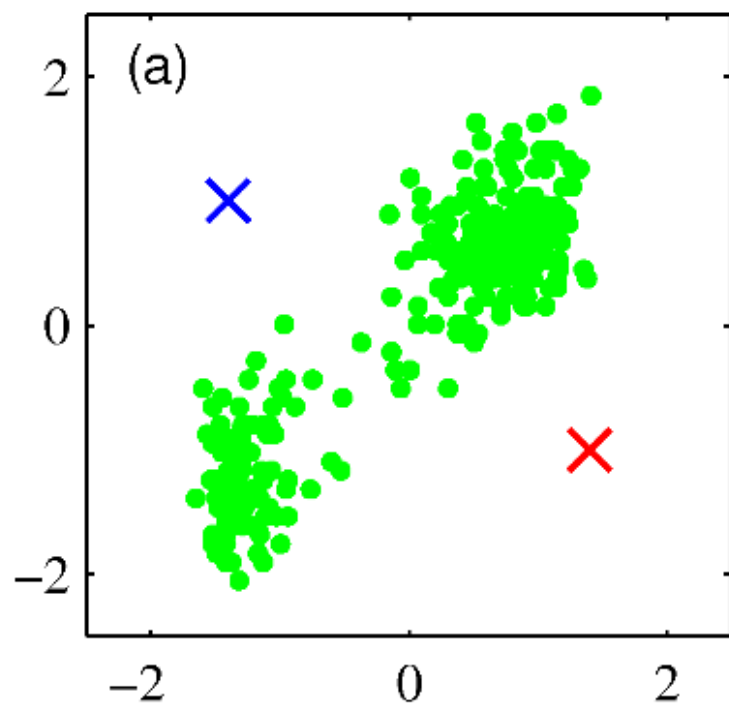


Old Faithful Geyser, Yellowstone

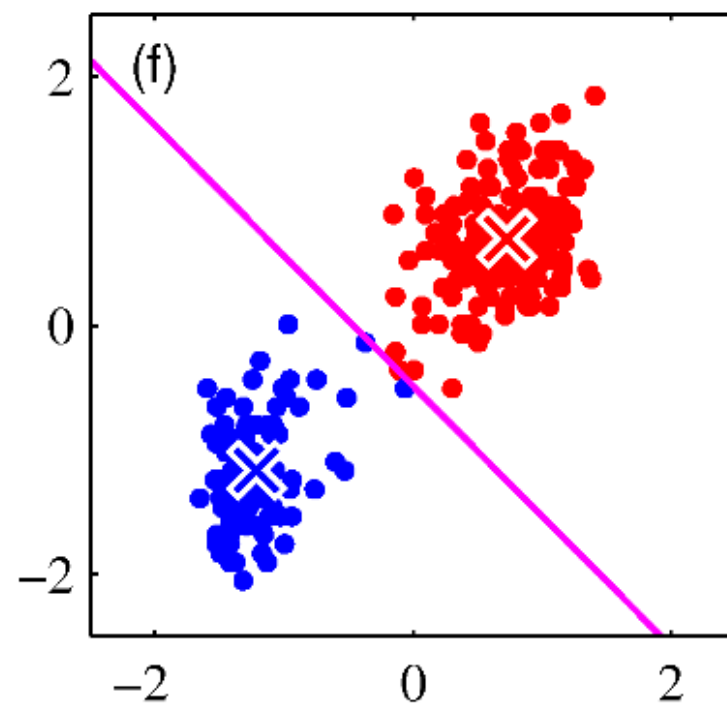
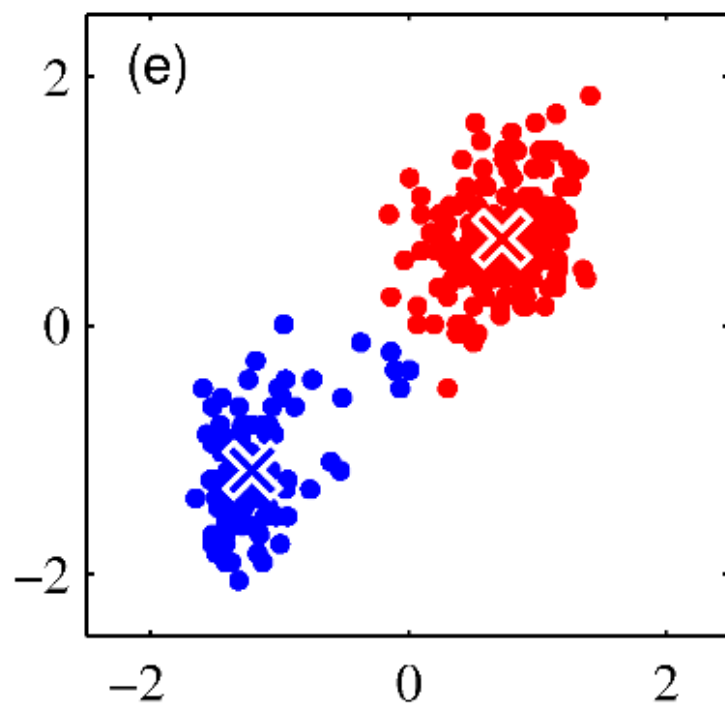
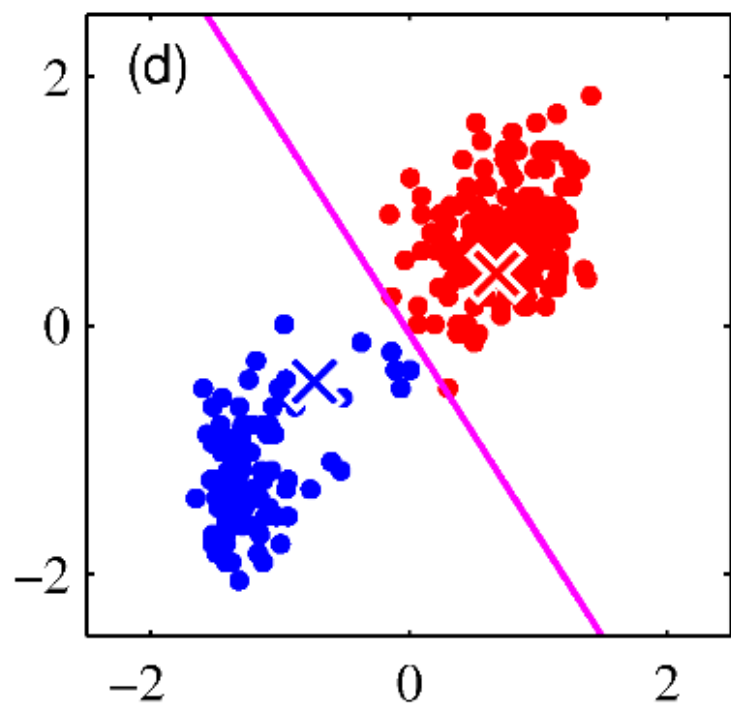


k-means Clustering

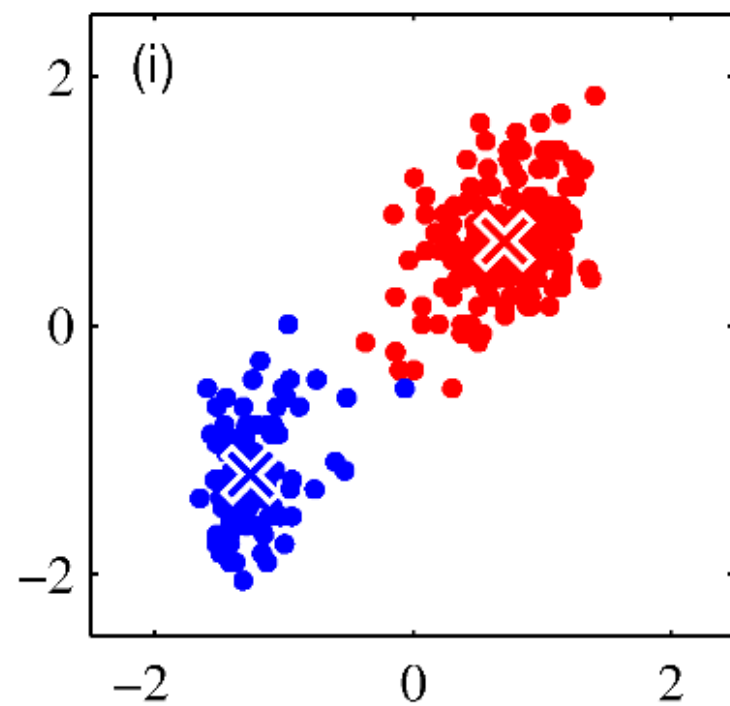
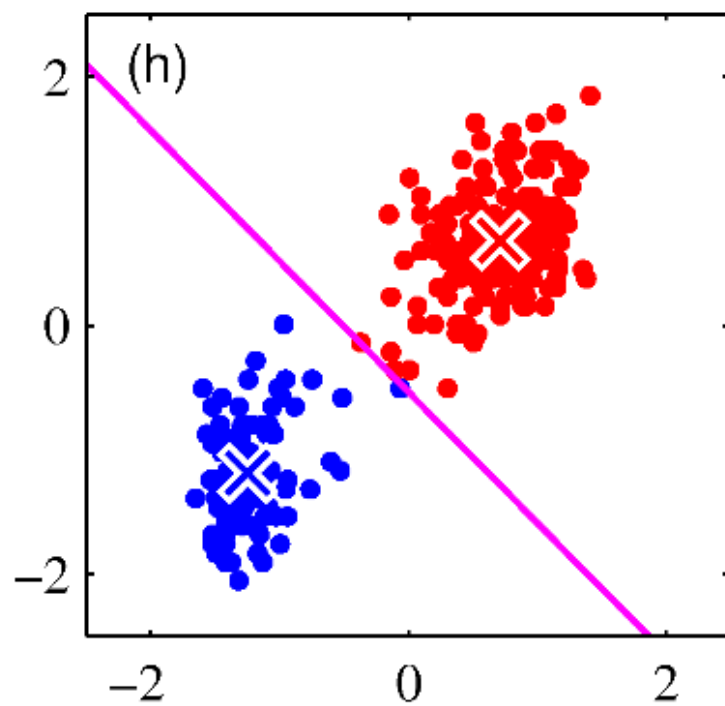
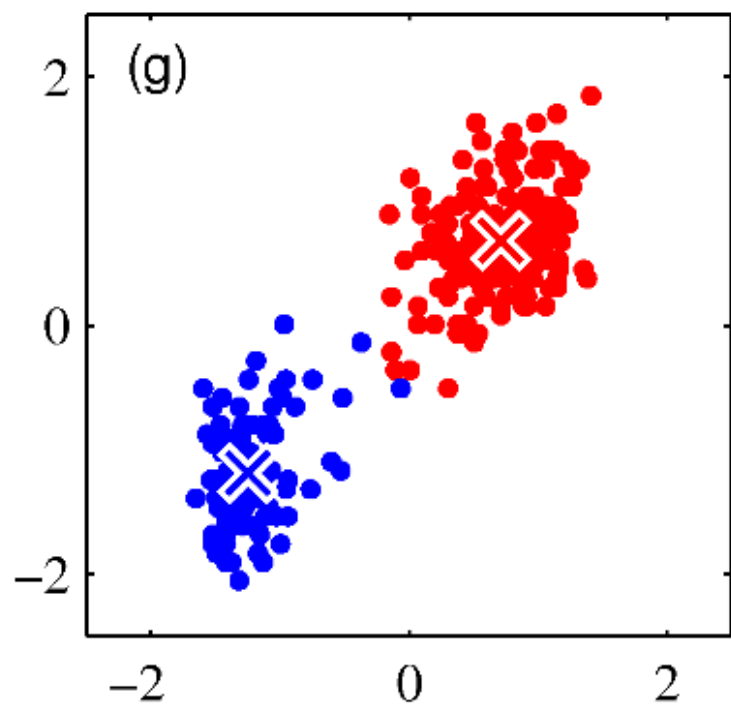
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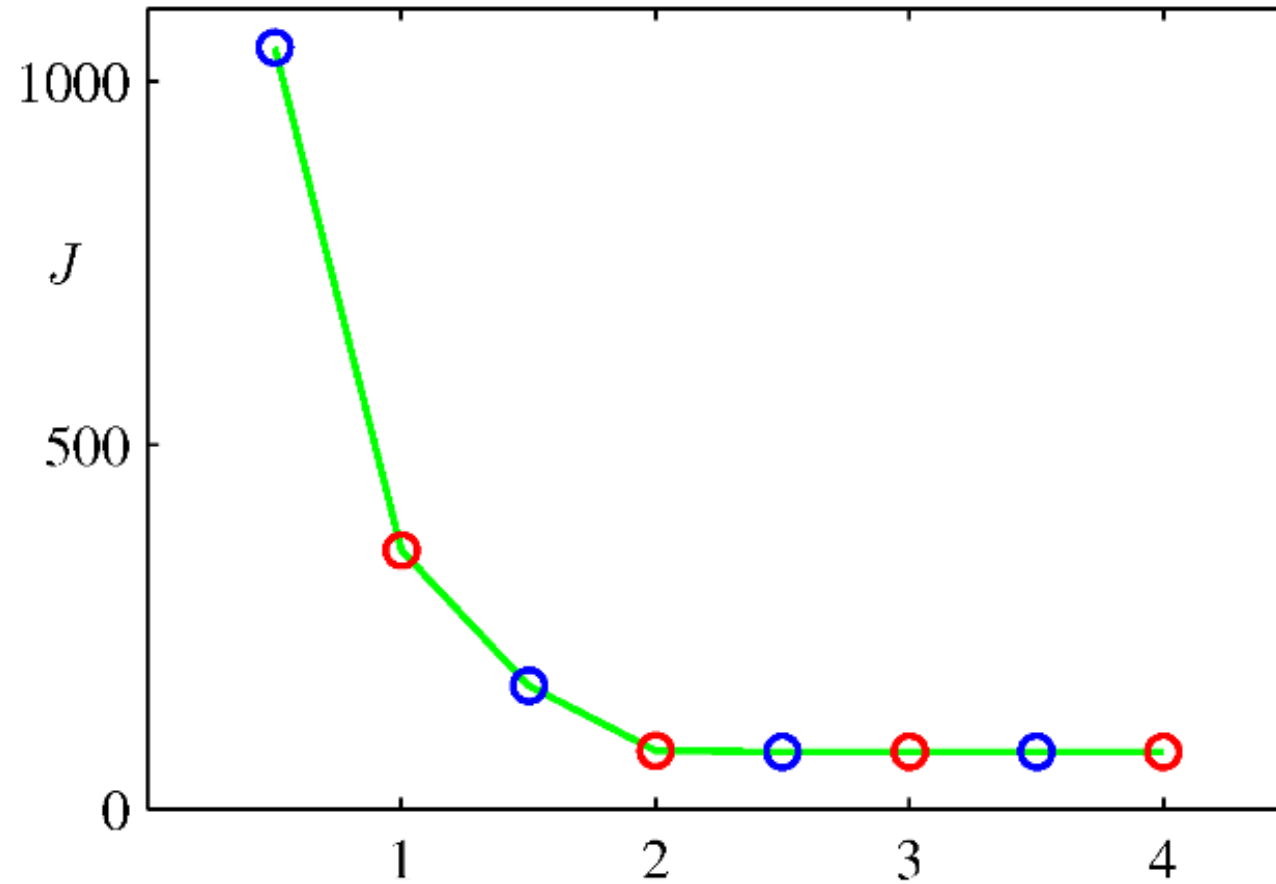
k-means Clustering



k-means Clustering



Objective function vs k-means iteration



Algorithm: k-means clustering

- Data $\mathcal{X} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N\} \in \mathbb{R}^D$
- Goal: Partition into K clusters $C = C_1, C_2, \dots, C_K$
- Cluster representatives (prototypes/codebook vectors)

$$V = \{\mu_1, \mu_2, \dots, \mu_K\}, \quad \mu_i \in \mathbb{R}^D$$

- Indicator variables describing cluster membership

$$r_{nk} \in \{0, 1\}, k = 1, \dots, K, \quad n = 1, \dots, N$$

- Objective function (Loss)

$$J = \sum_{n=1}^N \sum_{k=1}^K r_{nk} \|\mathbf{x}_n - \mu_k\|^2$$

Linear in r_{nk}

Quadratic in μ_k

Algorithm: k-means clustering

$$J = \sum_{n=1}^N \sum_{k=1}^K r_{nk} ||\mathbf{x}_n - \mu_k||^2$$

Repeat

- Alternating Optimization Strategy

- Fix μ_k and solve for r_{nk}

$$r_{nk} = \begin{cases} 1 & \text{if } k = \arg \min_j ||\mathbf{x}_n - \mu_j||^2 \\ 0 & \text{otherwise} \end{cases}$$

- Fix r_{nk} and solve for μ_k
 - Differentiate and solve

$$2 \sum_{n=1}^N r_{nk} (\mathbf{x}_n - \mu_k) = 0$$

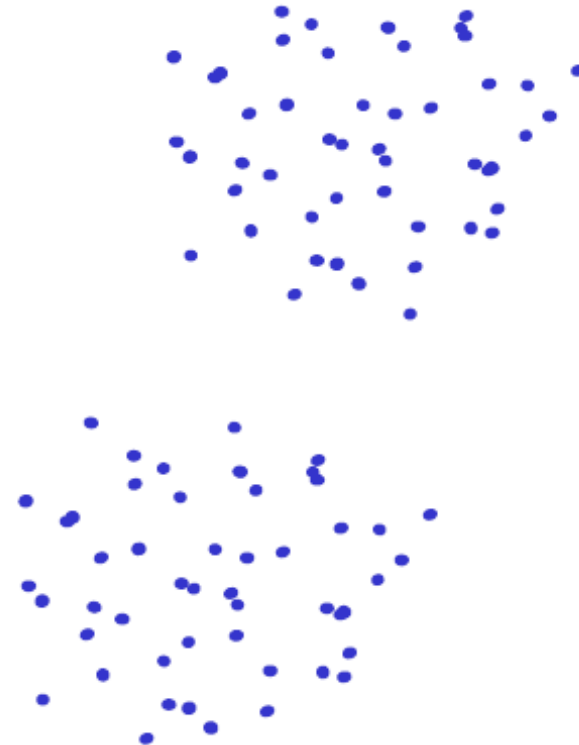
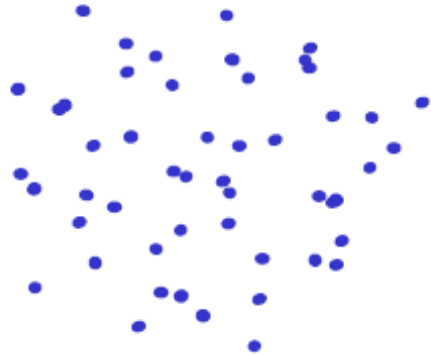
$$\mu_k = \frac{\sum_n r_{nk} \mathbf{x}_n}{\sum_n r_{nk}}$$

Variations of k-means

- Online k-means
 - Data coming in point by point
 - Sequential update of μ_k
- k-medoids
 - Mean may not coincide with a data point
 - k-medoids pick data points as cluster representatives and can use other distance functions
- Many other variants
 - See: Anil Jain, “Data clustering: 50 years beyond K-means”, Pattern Recognition Letters, Volume 31 Issue 8, Pages 651-666, Jun. 2010

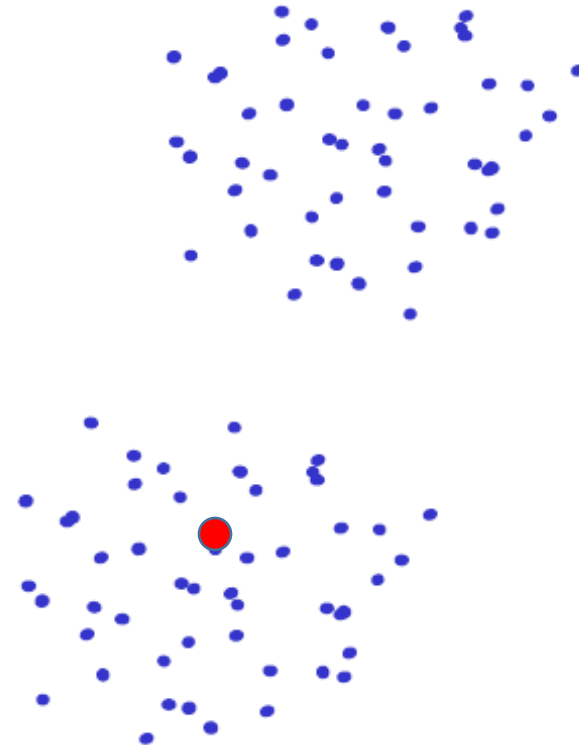
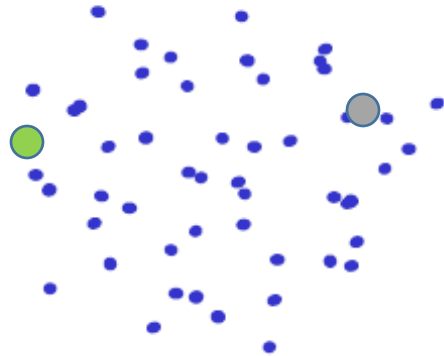
K-means

- Disadvantages
 - Dependent on initialization



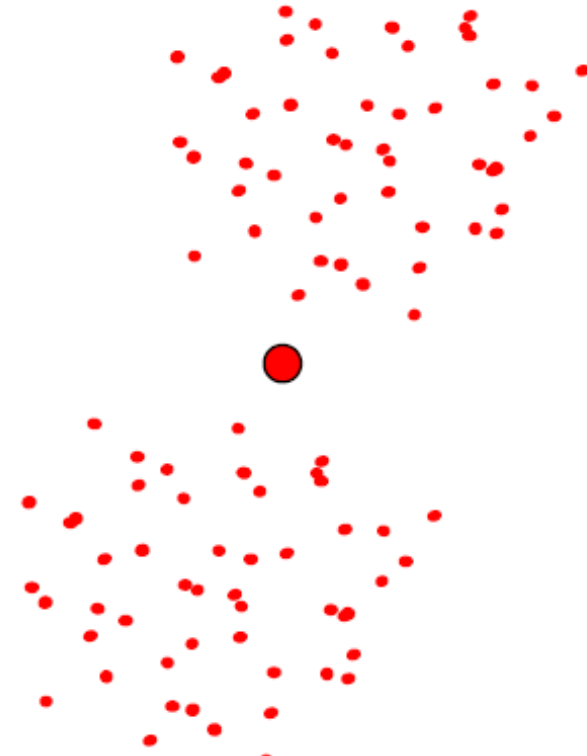
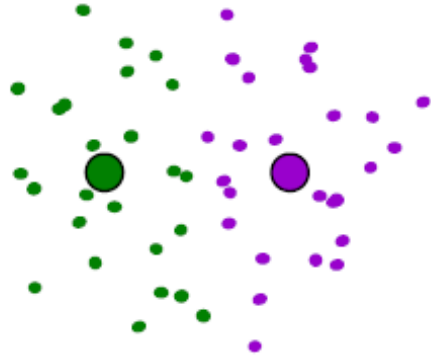
K-means

- Disadvantages
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K-means

- Disadvantages
 - Dependent on initialization

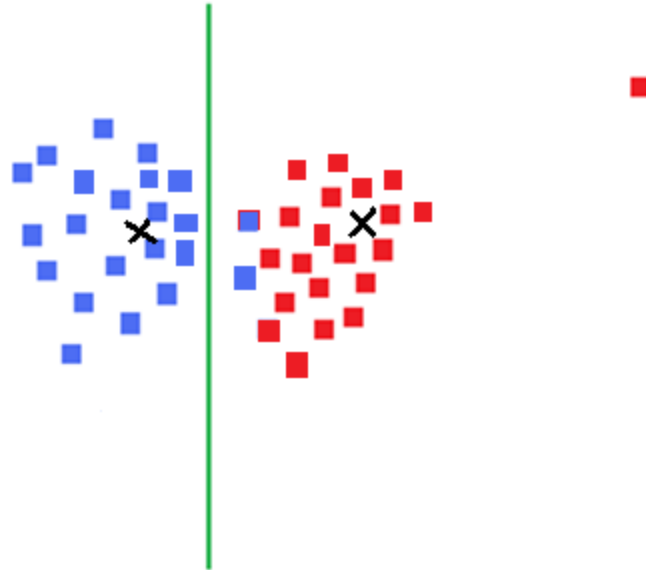


K-means

- Disadvantages
 - Dependent on initialization
 - Select random seeds with at least $D_{\min}$
 - Or, run the algorithm many times

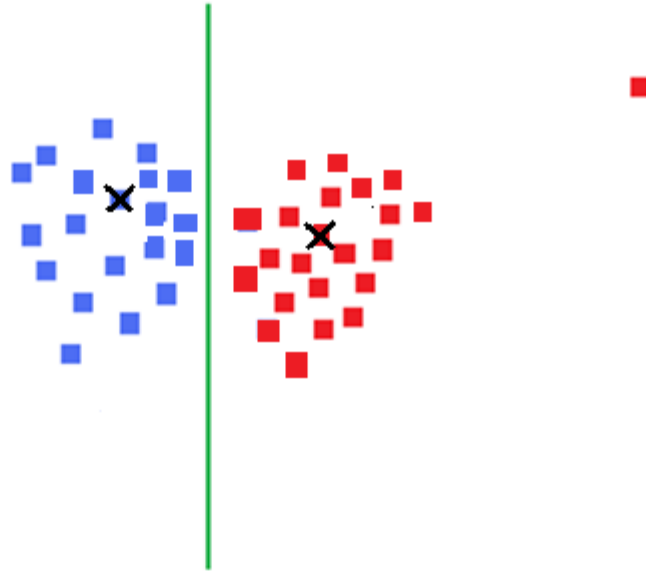
K-means

- Disadvantages
 - Dependent on initialization
 - Sensitive to outliers



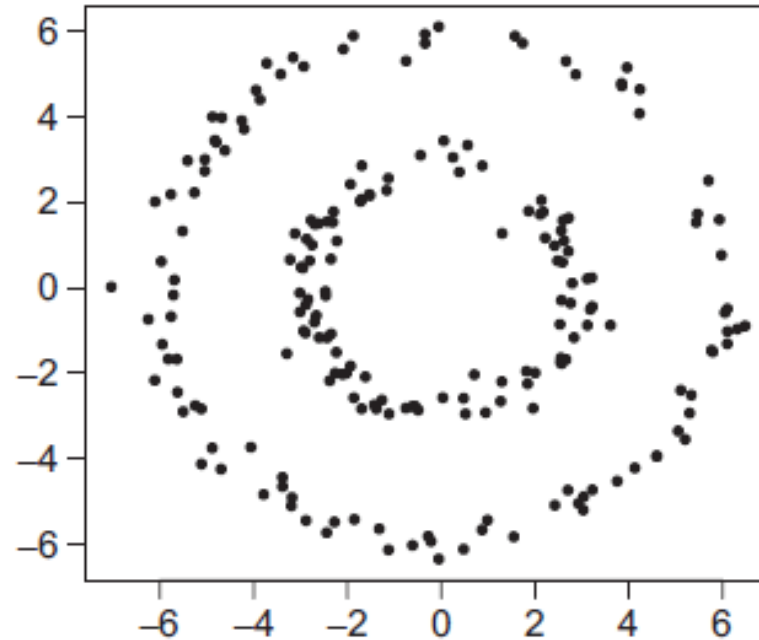
K-means

- Disadvantages
 - Dependent on initialization
 - Sensitive to outliers
 - Use K-medoids



K-means

- Disadvantages
 - Dependent on initialization
 - Sensitive to outliers (K-medoids)
 - Can deal only with convex clusters
 - Kernel trick/ Spectral Clustering

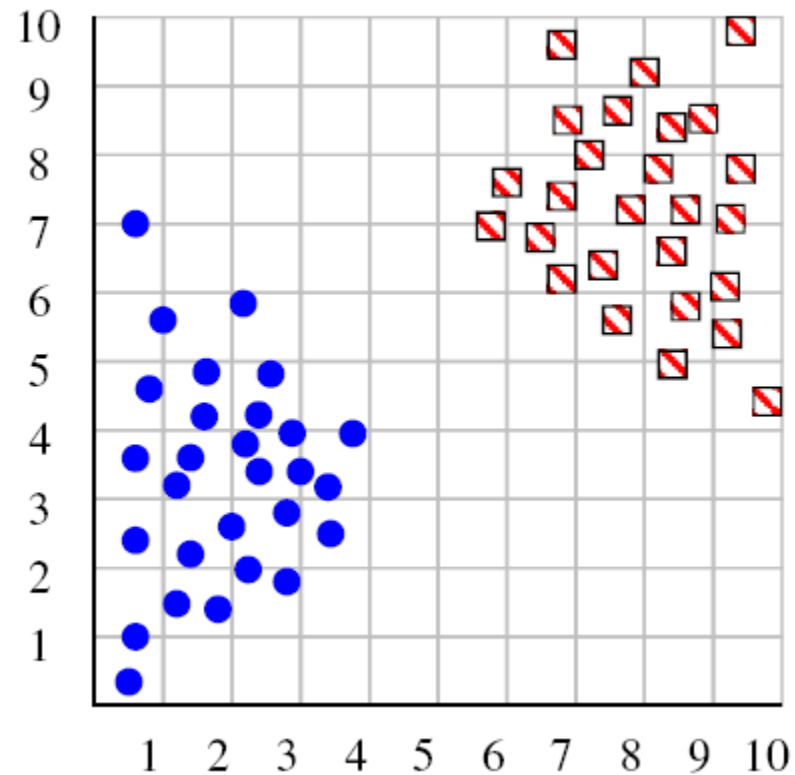


K-means

- Disadvantages
 - Dependent on initialization
 - Sensitive to outliers (K-medoids)
 - Can deal only with convex shaped clusters
 - Deciding K
 - Hard Assignment (sometimes a disadvantage)
 - A point can belong to one and only one cluster

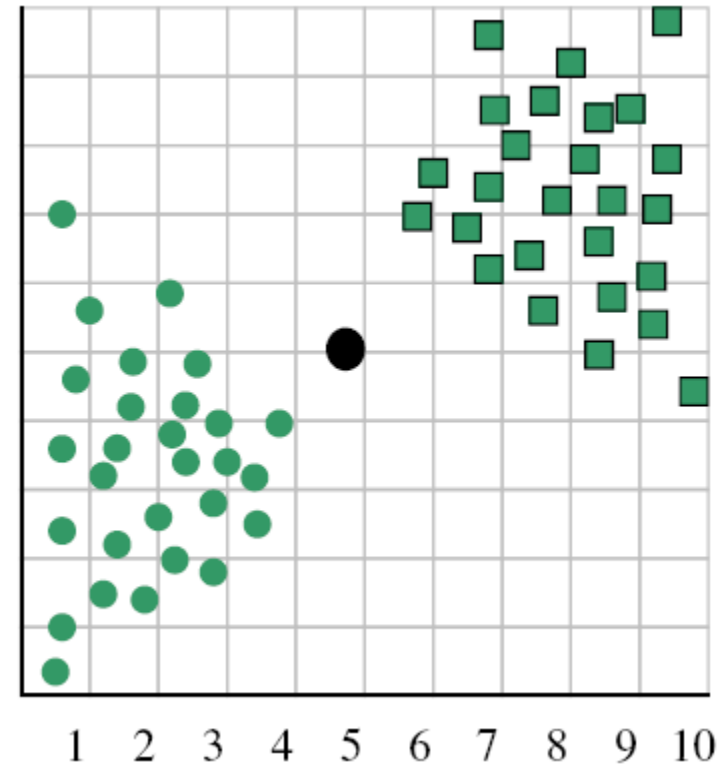
Deciding K

- Try a couple of K



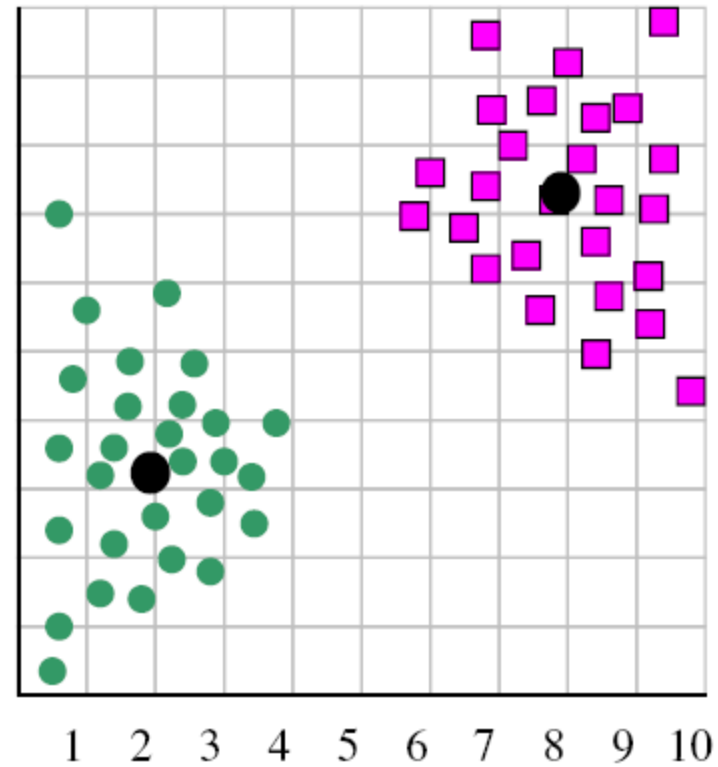
Deciding K

- When $k = 1$, the objective function is 873.0



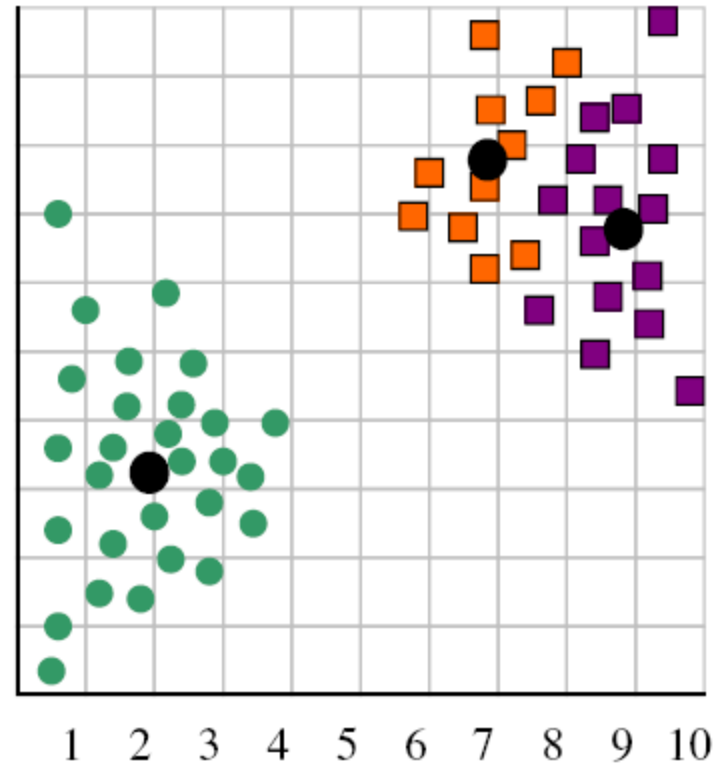
Deciding K

- When $k = 2$, the objective function is 173.1



Deciding K

- When $k = 3$, the objective function is 133.6



Deciding K

- We can plot objective function values for $k=1$ to 6
- The abrupt change at $k=2$ is highly suggestive of two clusters
- “knee finding” or “elbow finding”
- Note that the results are not always as clear cut as in this toy example

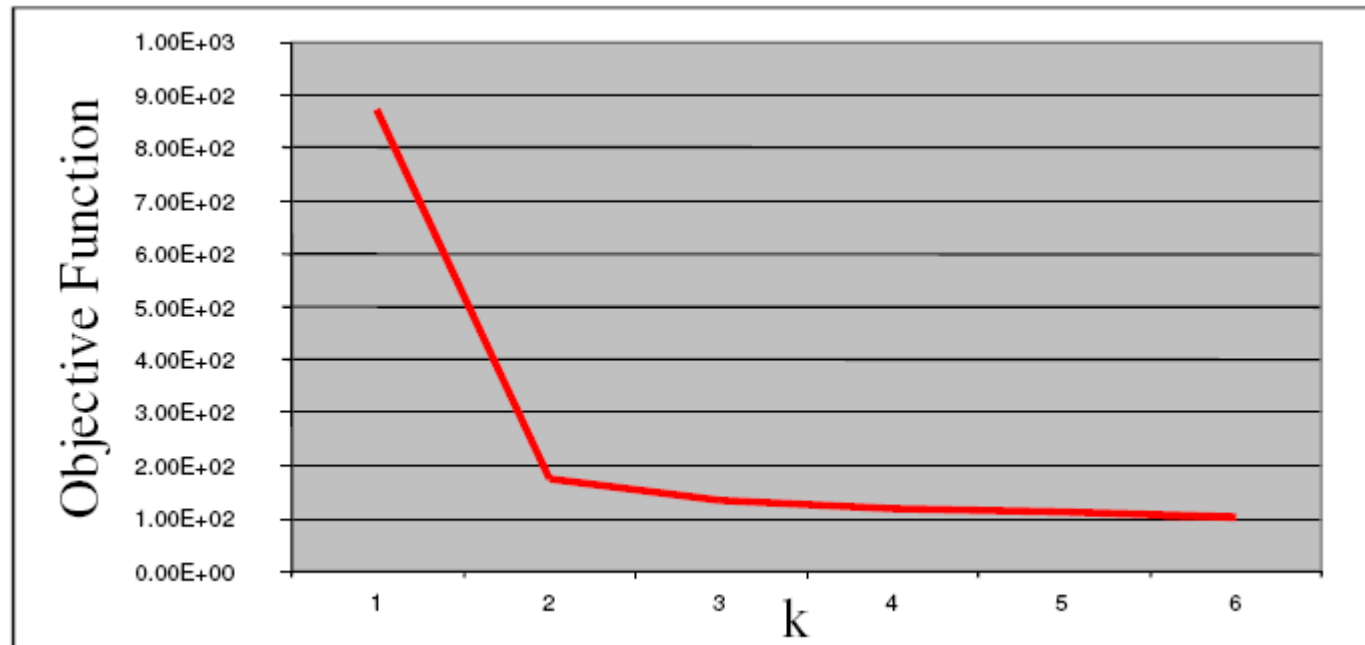
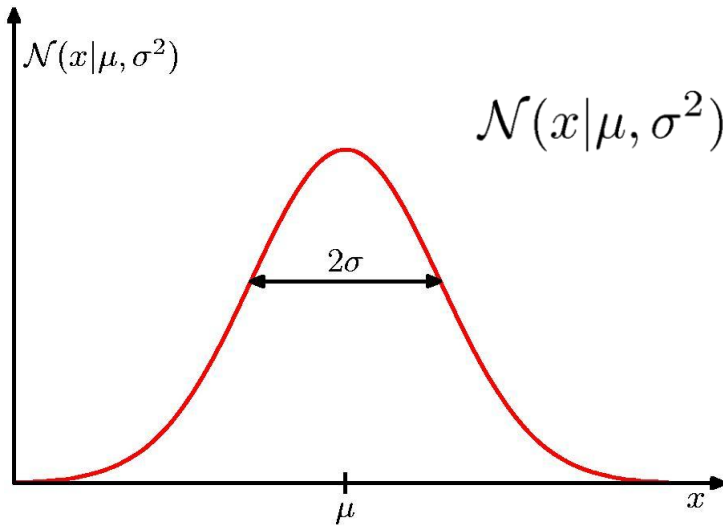


Image: Henry Lin

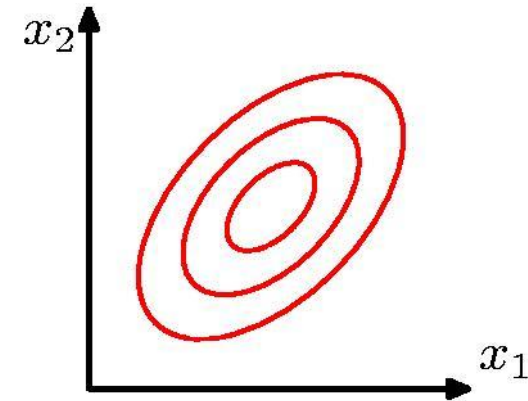
K-means

- Disadvantages
 - Dependent on initialization
 - Sensitive to outliers (K-medoids)
 - Can deal only with convex shaped clusters
 - Deciding K
 - Hard Assignment (sometimes a disadvantage)
 - A point can belong to one and only one cluster

The Gaussian Distribution



$$\mathcal{N}(x|\mu, \sigma^2) = \frac{1}{(2\pi\sigma^2)^{1/2}} \exp \left\{ -\frac{1}{2\sigma^2} (x - \mu)^2 \right\}$$



$$\mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{D/2}} \frac{1}{|\boldsymbol{\Sigma}|^{1/2}} \exp \left\{ -\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right\}$$

Moments of the Multivariate Gaussian

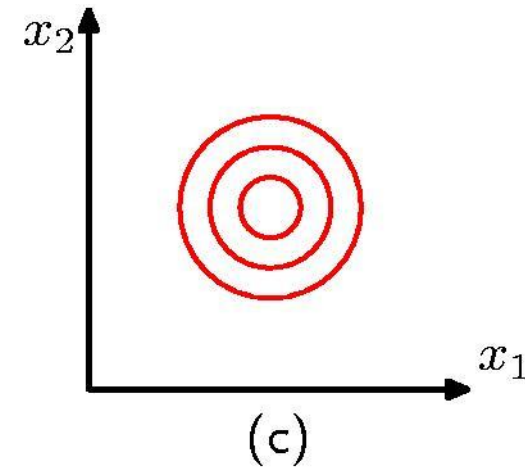
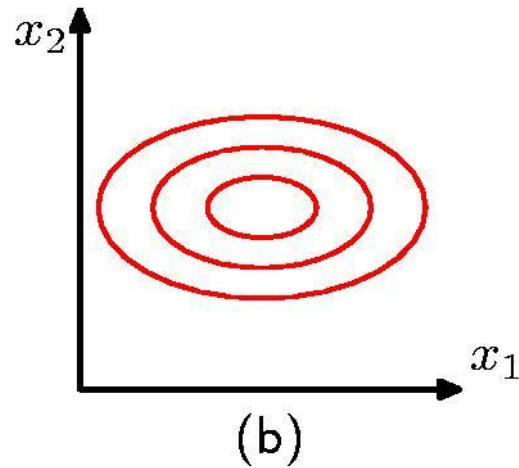
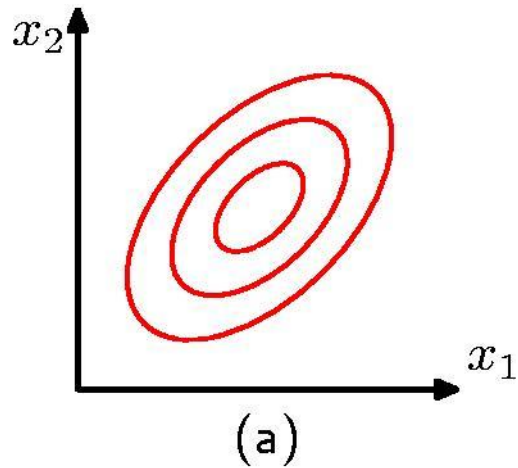
$$\begin{aligned}\mathbb{E}[\mathbf{x}] &= \frac{1}{(2\pi)^{D/2}} \frac{1}{|\boldsymbol{\Sigma}|^{1/2}} \int \exp \left\{ -\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu}) \right\} \mathbf{x} \, d\mathbf{x} \\ &= \frac{1}{(2\pi)^{D/2}} \frac{1}{|\boldsymbol{\Sigma}|^{1/2}} \int \exp \left\{ -\frac{1}{2}\mathbf{z}^T \boldsymbol{\Sigma}^{-1}\mathbf{z} \right\} (\mathbf{z} + \boldsymbol{\mu}) \, d\mathbf{z}\end{aligned}$$

$$\mathbb{E}[\mathbf{x}] = \boldsymbol{\mu}$$

Moments of the Multivariate Gaussian

$$\mathbb{E}[\mathbf{x}\mathbf{x}^T] = \boldsymbol{\mu}\boldsymbol{\mu}^T + \boldsymbol{\Sigma}$$

$$\text{cov}[\mathbf{x}] = \mathbb{E} [(\mathbf{x} - \mathbb{E}[\mathbf{x}])(\mathbf{x} - \mathbb{E}[\mathbf{x}])^T] = \boldsymbol{\Sigma}$$



Maximum Likelihood for the Gaussian

- Given i.i.d data $\mathbf{X} = (\mathbf{x}_1, \dots, \mathbf{x}_N)^T$, the likelihood function is a function of the parameters $(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and is given by

$$p(\mathbf{X}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \prod_{n=1}^N \mathcal{N}(\mathbf{x}_n|\boldsymbol{\mu}, \boldsymbol{\Sigma})$$

- In practice log-likelihood is convenient to optimize

$$\ln p(\mathbf{X}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) = -\frac{ND}{2} \ln(2\pi) - \frac{N}{2} \ln |\boldsymbol{\Sigma}| - \frac{1}{2} \sum_{n=1}^N (\mathbf{x}_n - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x}_n - \boldsymbol{\mu})$$

- Sufficient statistics
 - Statistics that we need to describe the entire distribution

$$\sum_{n=1}^N \mathbf{x}_n$$

$$\sum_{n=1}^N \mathbf{x}_n \mathbf{x}_n^T$$

Maximum Likelihood for the Gaussian

- Set the derivative of the log likelihood function to zero,

$$\frac{\partial}{\partial \boldsymbol{\mu}} \ln p(\mathbf{X} | \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \sum_{n=1}^N \boldsymbol{\Sigma}^{-1} (\mathbf{x}_n - \boldsymbol{\mu}) = 0$$

- and solve to obtain

$$\boldsymbol{\mu}_{\text{ML}} = \frac{1}{N} \sum_{n=1}^N \mathbf{x}_n.$$

- Similarly

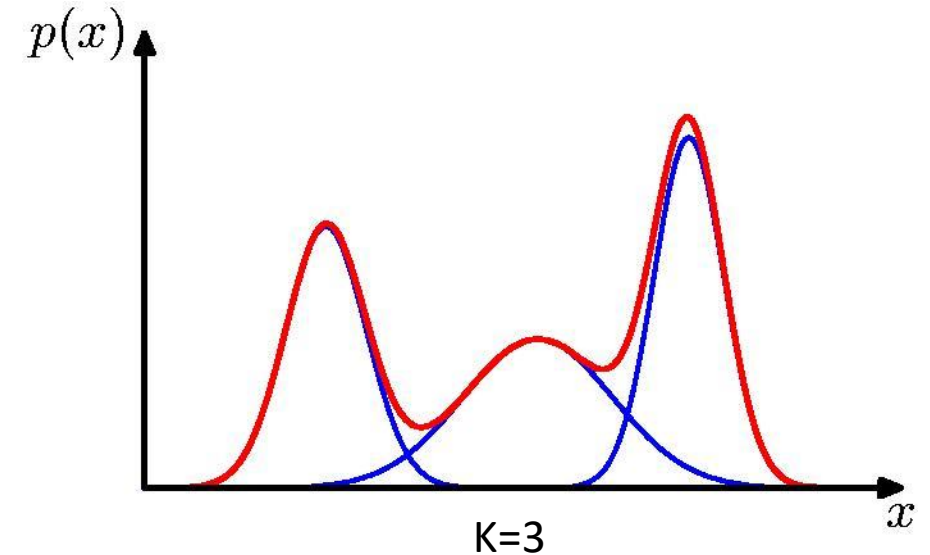
$$\boldsymbol{\Sigma}_{\text{ML}} = \frac{1}{N} \sum_{n=1}^N (\mathbf{x}_n - \boldsymbol{\mu}_{\text{ML}})(\mathbf{x}_n - \boldsymbol{\mu}_{\text{ML}})^{\text{T}}.$$

Mixtures of Gaussians

- Combine simple models into a complex model:

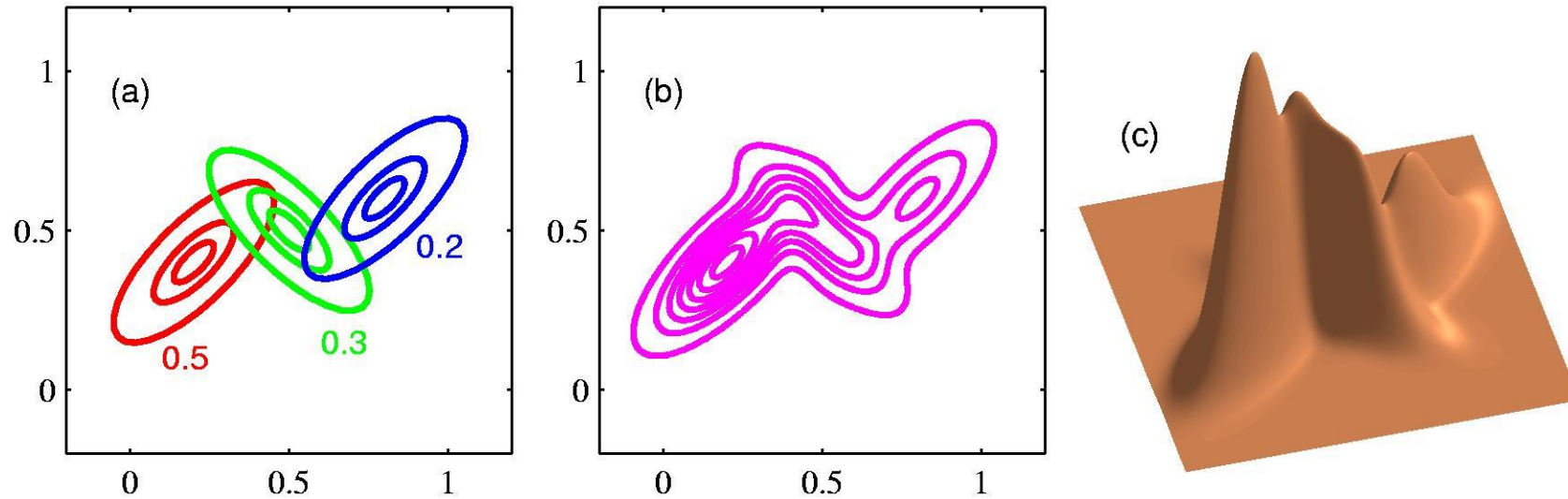
$$p(\mathbf{x}) = \sum_{k=1}^K p(\mathbf{x}|z = k)p(z = k) = \sum_{k=1}^K \underbrace{\pi_k}_{\text{Mixing coefficient}} \underbrace{\mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}_{\text{Component}}$$

- Mixing coefficients are the prior probabilities $p(z = k) = \pi_k$ of the k^{th} Gaussian component
- Posterior probabilities $p(z|\mathbf{x})$ also called '*responsibilities*'
 - the responsibility a component takes to 'explain' a point



$$\forall k : \pi_k \geq 0 \quad \sum_{k=1}^K \pi_k = 1$$
$$p(z = k) = \pi_k$$

Mixtures of Gaussians



- Also known as Gaussian Mixture Model (GMM)

Mixtures of Gaussians

- Determining parameters $\boldsymbol{\mu}$, $\boldsymbol{\Sigma}$, and π using maximum log likelihood

$$\ln p(\mathbf{X}|\boldsymbol{\pi}, \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \sum_{n=1}^N \ln \left\{ \underbrace{\sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}_{\text{Log of a sum; no closed form maximum.}} \right\}$$

Log of a sum; no closed form maximum.

- Difficult to maximize this function!

Mixtures of Gaussians

- Determining parameters μ , Σ , and π using maximum log likelihood

$$\ln p(\mathbf{X}|\pi, \mu, \Sigma) = \sum_{n=1}^N \ln \left\{ \underbrace{\sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x}_n | \mu_k, \Sigma_k)} \right\}$$

Log of a sum; no closed form maximum.

- Solution: use standard, iterative, numeric optimization methods or the *expectation maximization* (EM) algorithm.

Estimating the mean

$$\ln p(\mathbf{X}|\boldsymbol{\pi}, \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \sum_{n=1}^N \ln \left\{ \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) \right\}$$

- Differentiating wrt $\boldsymbol{\mu}_k$, we get

$$\sum_{n=1}^N \frac{\pi_k \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_{j=1}^K \pi_j \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)} \boldsymbol{\Sigma}^{-1} (\mathbf{x}_n - \boldsymbol{\mu}_k) = \sum_{n=1}^N \gamma_{nk} \boldsymbol{\Sigma}^{-1} (\mathbf{x}_n - \boldsymbol{\mu}_k) = 0$$

Not closed form any more

$$\boldsymbol{\mu}_k = \frac{1}{N_k} \sum_{n=1}^N \gamma_{nk} \mathbf{x}_n$$

$$N_k = \sum_{n=1}^N \gamma_{nk}$$

Estimating the covariance

$$\ln p(\mathbf{X}|\boldsymbol{\pi}, \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \sum_{n=1}^N \ln \left\{ \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) \right\}$$

- Differentiating wrt $\boldsymbol{\Sigma}_k$, we get

$$\boldsymbol{\Sigma}_k = \frac{1}{N_k} \sum_{n=1}^N \gamma_{nk} (\mathbf{x}_n - \boldsymbol{\mu}_k) (\mathbf{x}_n - \boldsymbol{\mu}_k)^\top$$

$$N_k = \sum_{n=1}^N \gamma_{nk}$$

Estimating the mixing coefficients

$$\ln p(\mathbf{X}|\boldsymbol{\pi}, \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \sum_{n=1}^N \ln \left\{ \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) \right\}$$

- For π_k , we have this constraint

$$\forall k : \pi_k \geq 0 \quad \sum_{k=1}^K \pi_k = 1$$

- So we use the Lagrange multiplier

$$\ln p(\mathbf{X}|\boldsymbol{\pi}, \boldsymbol{\mu}, \boldsymbol{\Sigma}) + \lambda \left(\sum_{k=1}^K \pi_k - 1 \right)$$

- Differentiating and equating to zero, we get

$$\sum_{n=1}^N \frac{\mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_{j=1}^K \pi_j \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)} + \lambda = 0$$

$$\pi_k = \frac{N_k}{N}$$

$$\gamma_{nk} = \frac{\pi_k \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_{j=1}^K \pi_j \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)}$$

$$N_k = \sum_{n=1}^N \gamma_{nk}$$

EM Algorithm

- Initialize the means, covariances and mixing coefficients
- E-step

$$\gamma_{nk} = \frac{\pi_k \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_{j=1}^K \pi_j \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)}$$

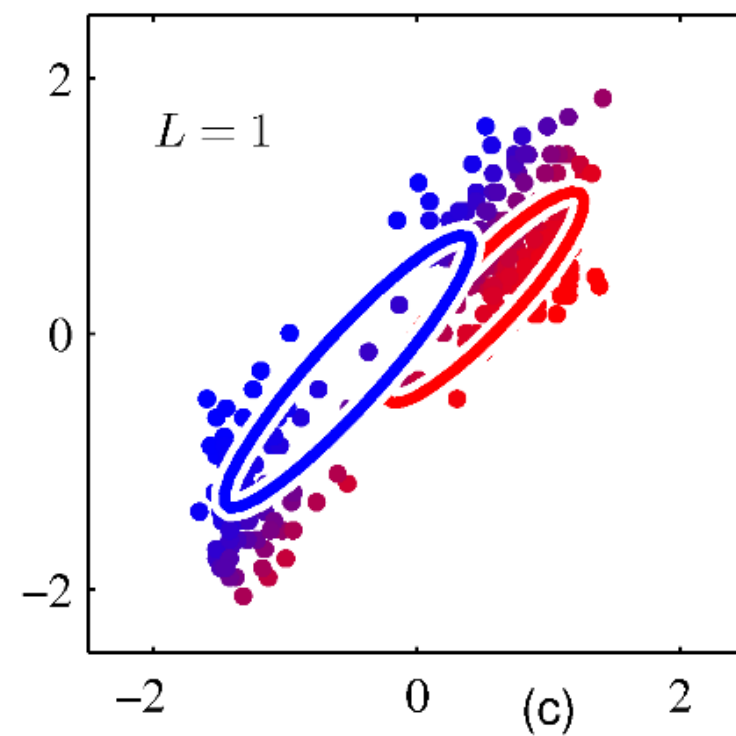
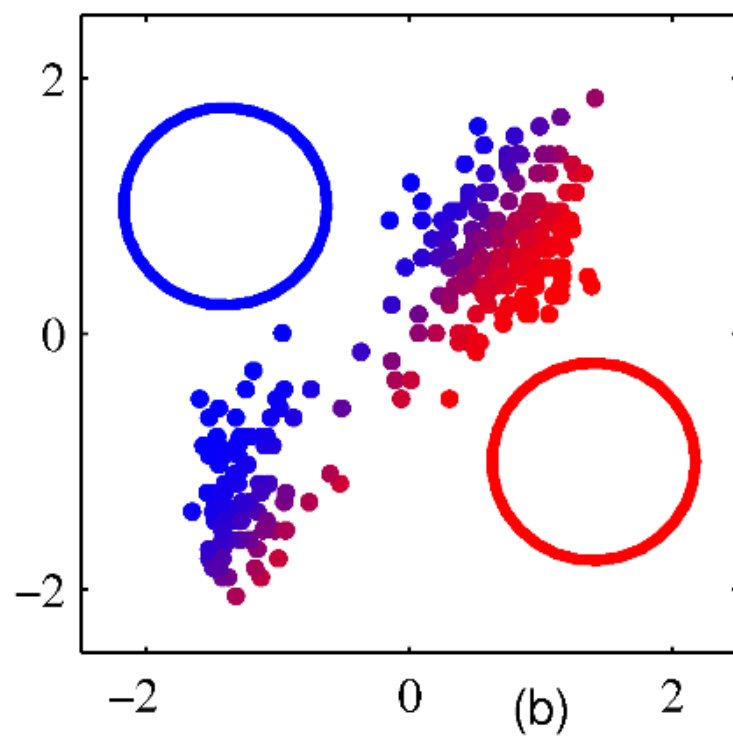
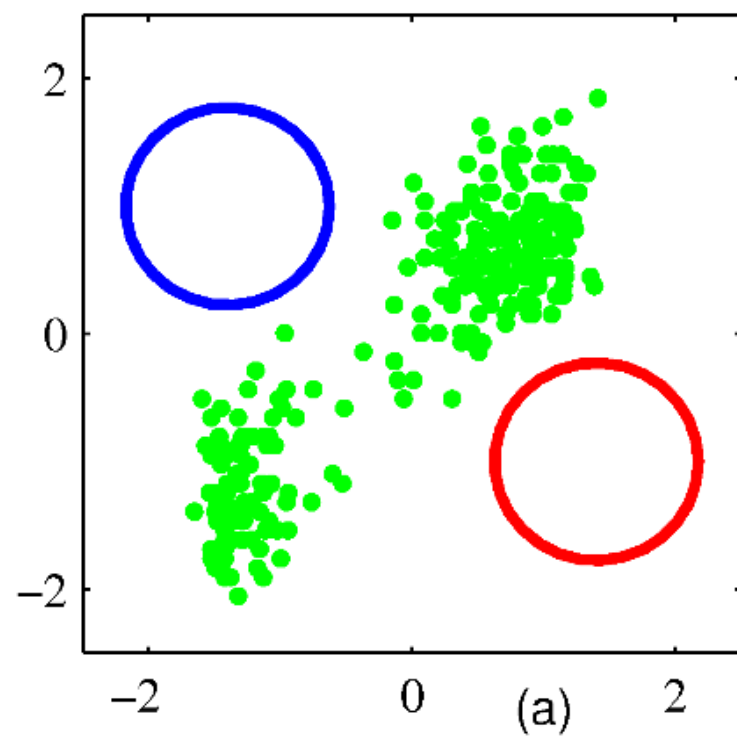
Repeat until
convergence

- M-step

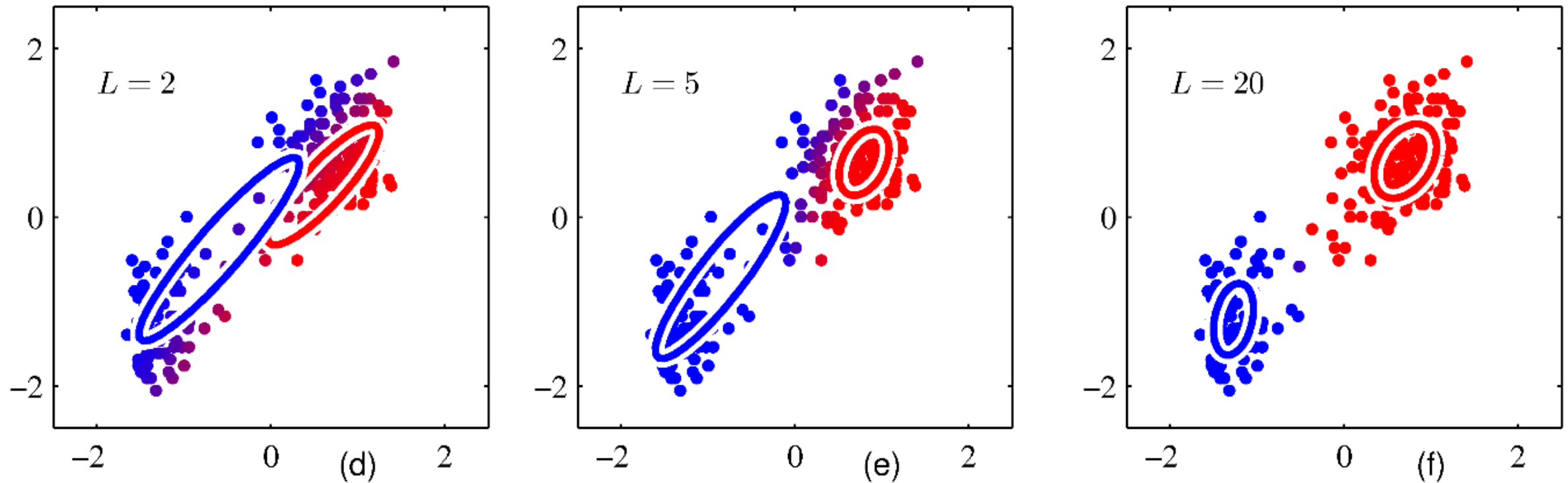
$$\begin{aligned}\boldsymbol{\mu}_k^{new} &= \frac{1}{N_k} \sum_{n=1}^N \gamma_{nk} \mathbf{x}_n \\ \boldsymbol{\Sigma}_k^{new} &= \frac{1}{N_k} \sum_{n=1}^N \gamma_{nk} (\mathbf{x}_n - \boldsymbol{\mu}_k^{new}) (\mathbf{x}_n - \boldsymbol{\mu}_k^{new})^\top \\ \pi_k^{new} &= \frac{N_k}{N}\end{aligned}$$

$$N_k = \sum_{n=1}^N \gamma_{nk}$$

Mixture of Gaussians on Old Faithful



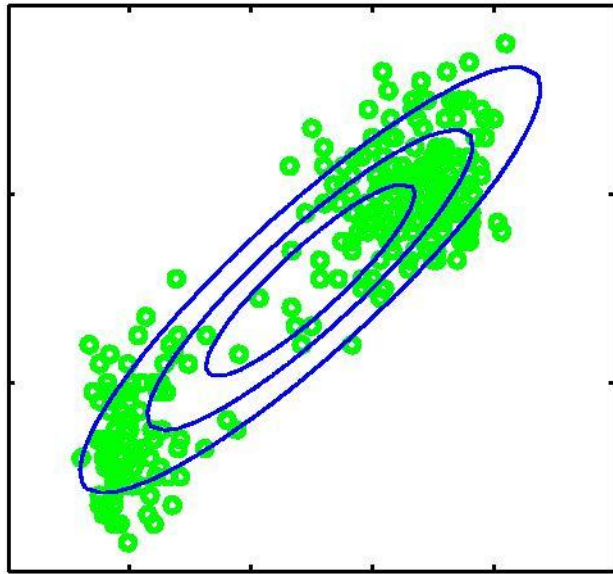
Mixture of Gaussians on Old Faithful



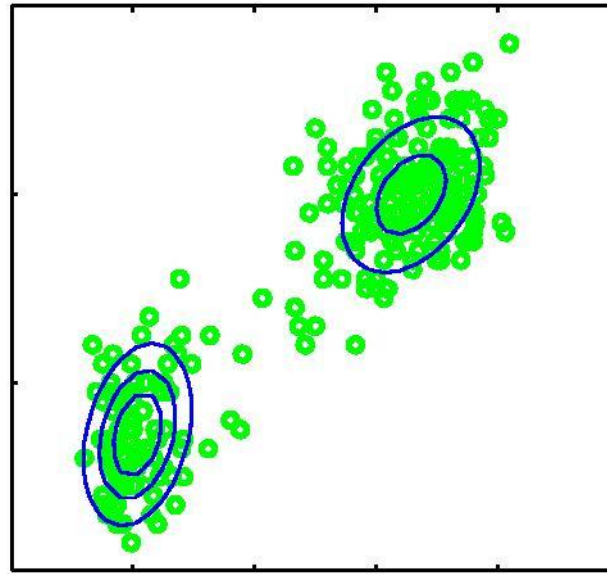
L is iteration number. In general EM takes much longer to converge as compared to k-means. In this example, recall that k-means converged in 3-4 iterations.

Mixtures of Gaussians

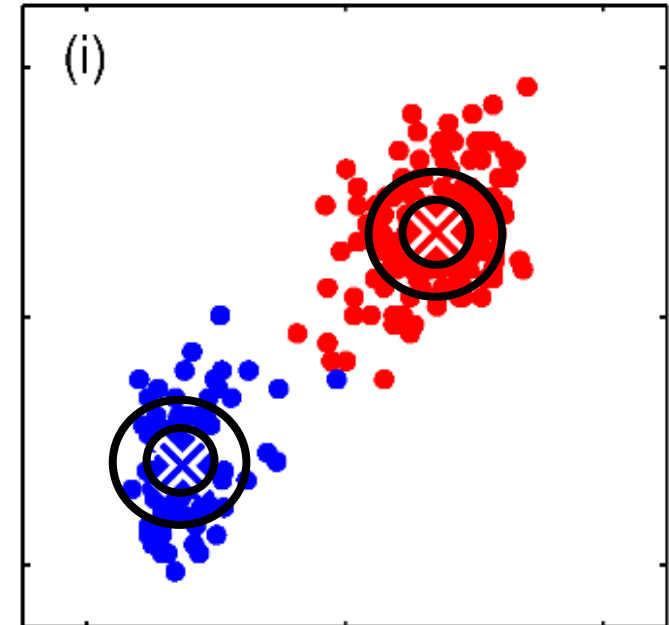
- Old Faithful data set



Single Gaussian



Mixture of two Gaussians



k-means: Two Gaussians with
Identity Covariance